

K1385 — TIER-INTEGRITY AUDIT of **data/bst_constants.json** (self-directed, follows K1384). ★ The 16/3 was NOT a one-off: Grace’s clean-repo gate (2026-08-02 + 2026-08-09) has systematically re-tiered over-claimed [D] constants DOWNWARD, concentrated in exactly the identification-heavy sectors (CKM/CP Wolfenstein, PMNS angles, nuclear magic numbers, cosmology fractions). That’s the survival-audit discipline (K1383) working at the DATA LAYER — healthy. ★ BUT two stragglers still PUBLISH [D] while flagged for demotion — const_015 (DM 16/3) and const_019 (Ω_b) — and the “68 Derived” headline is stale (lags the applied re-tierings). RULING: both stragglers → PARTIALLY DERIVED (adjudicated, Cal+Keeper delegated authority); the “DERIVED” label typo normalizes to “D”; no outreach cites a fixed Derived count until Grace’s gate closes + Cal cold-reads. Edits STAGED for the EOD data-sync (Grace’s lane) — I rule, I do not edit the file mid-session. Cal cold-reads.

Keeper (2026-08-11, self-directed after K1384. Grace found the 16/3 tier-inflation by accident while running the cell-count. That means the constants file can drift from the honest tier without direct audit — so I swept it. The gap is real but small and already mostly closed by Grace’s gate; I’m ruling the two stragglers and flagging the stale headline. Cal cold-reads. Nothing pushed.)

What I found (full tier survey of 197 constants)

Current file tiers: 68 D, 66 S, 46 I, 11 PD, 4 C, 1 SUPERSEDED, 1 “DERIVED” (label typo). 20 carry a review/re-tier/dedup flag.

★ The re-tiering is a PATTERN, and an honest one. Grace’s clean-repo gate has already re-tiered a cluster of [D]→lower, concentrated exactly where BST does <1% identification or asserted-count, not forward derivation: - CKM/CP mixing: Wolfenstein A ($D \rightarrow C$, rides open V_{cb} K1313), $\bar{\rho}$ ($D \rightarrow I$), $\bar{\eta}$ (CP-magnitude off, S). - PMNS: $\sin^2\theta_{12}$ ($D \rightarrow I$), $\sin^2\theta_{13}$ ($D \rightarrow I$) — “<1% but not a forward derivation.” - Nuclear magic numbers 4–7 (28/50/82/126): $D \rightarrow PD$ (deformed-magic, uncertified κ_{ls} deformation, K1132/K1133); κ_{ls} spin-orbit strength $D \rightarrow PD$. - Cosmology fractions: Ω_m (PD), Ω_Λ (PD, over-determination flag), DM 16/3 (flagged → PD), Ω_b (flagged → PD). - Confirmed-honest (no change): H_0 (I — correctly inherited from CAMB inputs; “calibrate both directions, this one passes”).

This is the discipline working, not a scandal. The walk-back rate is high at the data layer too (K1383’s thesis holds one level down). The re-tierings are target-innocent and land in the sectors we’d predict. The *only* real defect is that the headline count lags the honest re-tiering.

★ RULING — the two stragglers (still publishing [D], flagged for demotion)

1. const_015 — DM/baryon 16/3 → PARTIALLY DERIVED. Adjudicated in K1384: the count $2^{(2-\text{rank})}$ is asserted (channel-capacity story), not derived by tiling the boundary; 16/3 is the nearest $2^k/N_c$ to obs (mild form-selection). $D \rightarrow PD$.
 2. const_019 — $\Omega_b = 18/361 \rightarrow$ PARTIALLY DERIVED. Downstream of Ω_m (PD) and the 16/3 (now PD); a downstream quantity cannot out-rank its weakest input (monotonicity). $D \rightarrow PD$.
- Authority: tier adjudication is delegated to Cal+Keeper (feedback: audit-chain governance; Casey override). These two are no longer “pending Casey/Cal” — they are RULED. Cal cold-reads to co-sign.

- Label hygiene: the lone "tier": "DERIVED" entry normalizes to "tier": "D" (typo; it inflates no claim but breaks any automated tier-count).

★ The governance consequence (what this protects)

- The “68 Derived constants” figure is STALE and must not be quoted externally as-is. After the two stragglers apply and the label normalizes, the honest Derived count is $\approx 66-67$, and it is *still under active downward revision* by Grace’s gate (dedup of duplicate CKM forms is queued, not closed). Standing: no outreach, paper, or README cites a fixed “N Derived” count until Grace’s clean-repo gate closes AND Cal cold-reads the final ledger. (This is the K1384/K1383 principle: the artifact’s honesty must be audited directly, not assumed from the K-ruling chain.)
- The gravity arc’s PD-floor (K1384) is now consistent with the data layer once const_015 applies — the two 16/3 entries and the gravity same-object claim all sit at PD, honestly, until the forward cell-count promotes them.
- Duplicate CKM/CP forms remain flagged-not-closed (A: $N_{c^2}/(2C_2-1)=9/11$ vs n_C/C_2 ; $\bar{\eta}$: $1/(2\sqrt{2})$ vs $n_C/(\text{rank-g})$). Dedup is Grace’s lane; the tier is already ruled (C/I/S). Not load-bearing for the current arc — noted so it isn’t forgotten.

Staged edits (for the EOD data-sync — Grace’s lane, zero-judgment)

```
const_015: "tier": "D"           → "tier": "PD"   (K1384/K1385; 16/3 count asserted)
const_019: "tier": "D"           → "tier": "PD"   (downstream inherit, monotonicity)
<the one entry>: "tier":"DERIVED" → "tier":"D"   (label typo normalize)
```

Then recompute the D/PD/I/C/S headline and re-sync the board/README/CLAUDE.md counters. I rule and stage; I do not edit the constants file mid-session (it feeds counters that sync to root docs — unilateral mid-session edits desync the board, which is exactly the process discipline Keeper enforces).

Why I ran this (standing value)

K1383 proved the *process* is honest. K1384 showed the *artifact* (constants file) can still drift from the honest tier — Grace caught the 16/3 by accident, not by audit. The lesson: the data layer needs its own periodic tier-integrity pass, because it’s what `verify_bst.py` and every outreach number are read from — the K-ruling chain’s honesty does not automatically propagate to the published artifact. Grace’s clean-repo gate is that pass and it’s working; my job is to (a) adjudicate the flagged tiers (done: two stragglers → PD), (b) hold the standing that the headline count is not quoted until the gate closes. Standing: re-run a constants-file tier-integrity sweep before any outreach that cites derived-count figures.

— Keeper, K1385, TIER-INTEGRITY AUDIT 2026-08-11. Swept all 197 constants: 68 D / 66 S / 46 I / 11 PD / 4 C / 1 SUPERSEDED / 1 “DERIVED”(typo). FINDING: Grace’s clean-repo gate (08-02 + 08-09) has systematically re-tiered over-claimed [D]→lower in the identification-heavy sectors (CKM/CP Wolfenstein $A \rightarrow C/\bar{\rho} \rightarrow I/\bar{\eta} \rightarrow S$, PMNS $\theta_{12}+\theta_{13} \rightarrow I$, nuclear magic 4-7 + $\kappa_{ls} \rightarrow PD$, cosmology fractions $\Omega_m/\Omega_\Lambda/\Omega_b/DM \rightarrow PD$) — HONEST, the survival-audit discipline (K1383) at the data layer; H_0 confirmed-honest (no change). Defect = the headline count LAGS the applied re-tierings. RULING: two stragglers still publishing [D] → PARTIALLY DERIVED — const_015 (DM 16/3, count asserted, K1384) and const_019 (Ω_b , downstream monotonicity); “DERIVED” label typo → “D”. Cal+Keeper delegated authority; no longer pending; Cal cold-reads. CONSEQUENCE: the “68 Derived” figure is STALE (honest $\approx 66-67$, still revising) — STANDING: no outreach cites a fixed Derived count until Grace’s gate closes + Cal cold-reads final ledger. Gravity PD-floor now data-consistent once const_015 applies. Duplicate CKM forms flagged-not-closed (Grace lane, tier already ruled). Edits STAGED for EOD data-sync (Grace’s lane); Keeper rules+stages, does NOT edit mid-session (counter-sync discipline). Standing: run a constants-file tier-integrity sweep before any derived-count outreach. Cal cold-reads. Nothing pushed.